ABRAHAM TEKLU

PARTICLE PHYSICS PH.D.

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LINKS

https://abrahamteklu.com/, https://www.linkedin.com/in/a-teklu/ https://lisea.org/

PROFILE

Particle Physics Ph.D. with experience in programming for data analysis. Used both Python and C++ for 9+ years for analysis on large data sets. Experience with presenting and communicating my ideas. As part of a team, created and executed relevant physics studies which led to new measurements of the neutron cross-section on scintillator.

EDUCATION

Stony Brook University, 2018
Doctor of Philosophy — Physics, Stony Brook

- GPA: 3.67
- Awards: Turner Fellowship, Lourie Fellowship

Oregon State University, 2014 to 2018
Bachelor of Science — Physics, Corvallis

- GPA: 3.80
- Awards: Academic Achievement Award, Nicodemus Scholar Award, Ken Krane Scholar Award

TECHNICAL PROFICIENCIES

Technical Skills:	Python, C++, ROOT, Linux Bash, Docker, Singularity, Matlab, HTML
Quantitative Skills:	Linear Algebra, Calculus, Statistics
Team Skills:	Communication in Large Collaborations, Presentation Experience
Research Skills:	Problem Solving, Data Analysis, Data Parsing, Algorithmic Programming

PROFESSIONAL EXPERIENCE

Graduate Research Assistant
Stony Brook University *Aug 2018 to Present*

Neutron Reconstruction for T2K [Super-FGD] (December 2021 – Present) *ROOT, C++, Python*

- T2K is a large collaboration of scientists, postdocs, and graduate students from around the world.
- Working on a team to develop data analysis tools to select neutrons in neutrino events.
- My current work improves neutrino energy calculations and the oscillation analysis fit, the main goal of T2K.

Construction of the Super-FGD for the T2K Near Detector Upgrade (October – December 2022)

- Worked with 20 people from across the world to assemble a detector made of 2 million scintillator cubes.
- Assembly took place at JPARC, a high-intensity proton accelerator facility located in Tokai, Japan.
- Completed the project and delivered the detector a month before the deadline.

T2K Oscillation Analysis in the MaCh3 Group (January – December 2021) *ROOT, C++, Docker, Singularity*

- Built a Docker container that enabled the first-ever joint collaboration oscillation analysis fit between T2K (with 500+ members in 12 countries) and NOvA (with 250+ members in 8 countries).

Novel 3D Scintillator Detector Prototypes (August 2019 – Present) *ROOT, C++, Python*

- Created and executed relevant physics studies that led to new measurements of the neutron cross-section and doubled the energy range of the existing neutron cross-section measurements on scintillator.
- Worked with 61 researchers from 19 universities on the prototype construction at CERN (Geneva, Switzerland) and Stony Brook University (Stony Brook, NY). Led a data-taking team in Los Alamos National Lab (Los Alamos, NM) and was a major contributor to the analysis.
- The work produced 3 co-authored publications and an APS April Meeting Talk.

LISEA — Programming Bootcamp, Feb 2023 to May 2023

- Created a 6-week data science curriculum for a small group of disenfranchised minorities.
- Lectured and have 1-on-1 sessions to improve programming skill and help with assignments.

Visiting Research Assistant, CERN, Jun 2018 to Aug 2018

- Helped assemble and test the Super-FGD Prototype, the world's first detector that can do event-by-event neutron kinematic reconstruction. Ran the first tests with a charged particle beam on a detector of this type.

SULI Internship, General Atomics DIII-D Tokamak Fusion Reactor, Jun 2017 to Aug 2017

- Worked with 3 staff scientists at DIII-D, to model plasma in the DIII-D Tokamak Fusion Reactor.
- Contributed to a co-authored publication, and presented a poster at the APS DPH Meeting. Also contributed to 4 other co-authored talks and posters.

CIERA REU, Northwestern University, Jun 2016 to Aug 2016

- Found an approximate numerical solution for an unstable region of the Holling-Tanner model computationally.
- Presented findings in a poster at the American Astronomical Society (AAS) and made a website.
  https://cierareu.northwestern.edu/REU2016websites/Teklu/index.html

Undergraduate Research, OSU, Jan 2016 to Jan 2018

- Modeled antineutrino interactions for the MINERvA detector at Fermilab in Batavia, Illinois.
- Analysis was included in a co-authored publication by the MINERvA collaboration.